Section 2 - Demand Forecasting for Childcare Places

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Problem Statement: Where Should Future Childcare Capacity Be Built?

Hypotheses

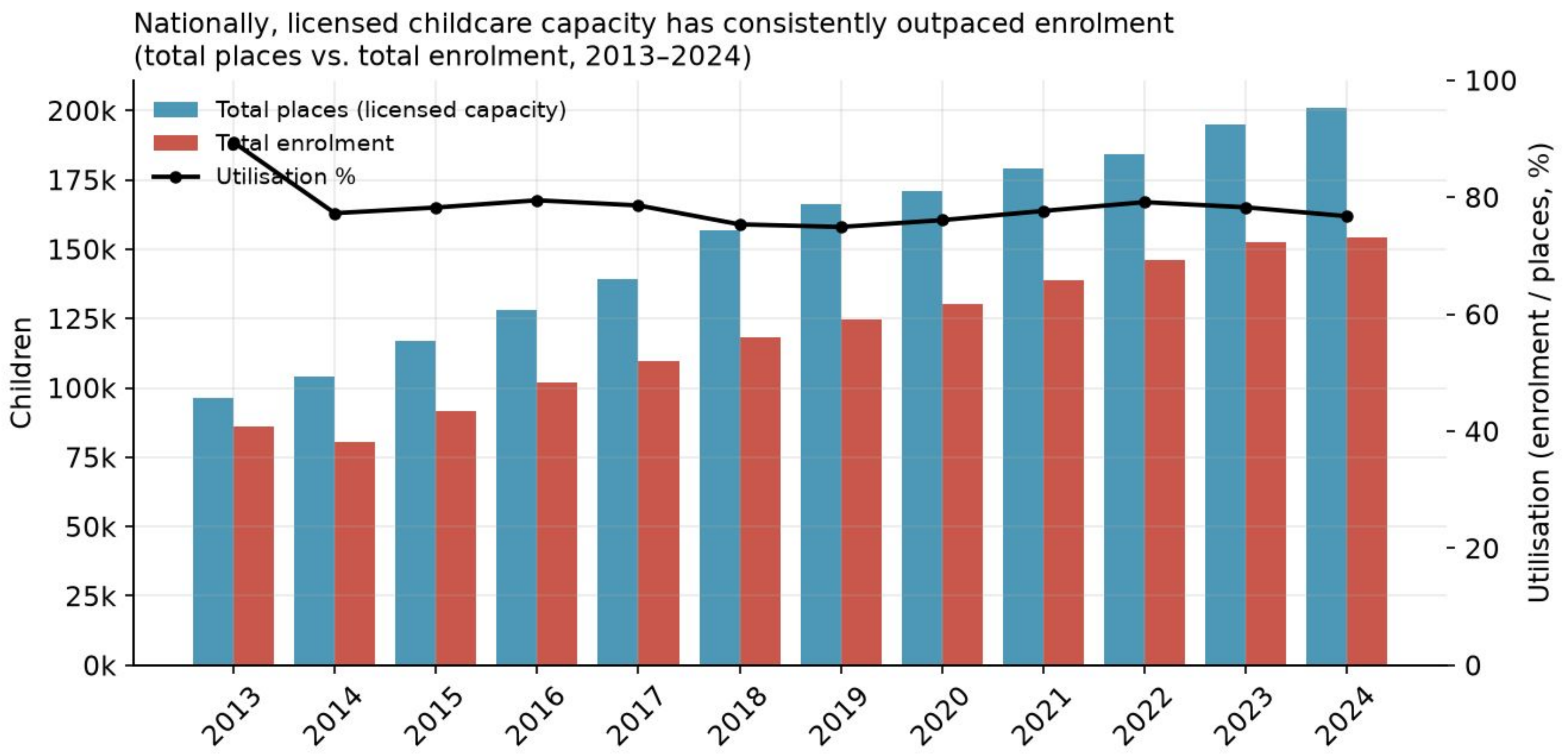
The analysis is built around three testable hypotheses, each evaluated against the evidence in Section 8.

If these hypotheses hold, the recommendations naturally separate into two strategies: **build new capacity** in high-growth BTO areas and **relocate or right-size capacity** in mature estates with declining demand.

#	Hypothesis	Supporting Evidence
H1	The childcare shortage is driven by spatial misallocation, not an overall national capacity shortage.	Large variation in demand–supply gaps across subzones despite adequate national capacity.
H2	BTO completions are a leading indicator of future childcare shortages.	Subzones with large BTO pipelines experience the fastest growth in childcare demand and the largest projected shortages.
H3	Falling fertility will reduce demand in mature, non-BTO estates.	Mature estates with limited BTO activity show stable or declining child populations, making them potential candidates for capacity reallocation.

FRAMING THE QUESTION

Is Shortage due to misallocation or overall capacity shortage?



1. National childcare utilisation has remained below ~80%, suggesting Singapore's challenge is not an overall shortage of childcare places.

2. The issue is likely a geographic mismatch between where childcare capacity exists and where demand is concentrated, shifting the focus from building more centres to placing them in the right locations.

3. This validates that hypothesis by identifying regional demand–supply gaps and assessing whether observed cases (*The Big Read* illustrates it with named estates (Punggol acute, Tiong Bahru comfortable)) are reflected in the data.

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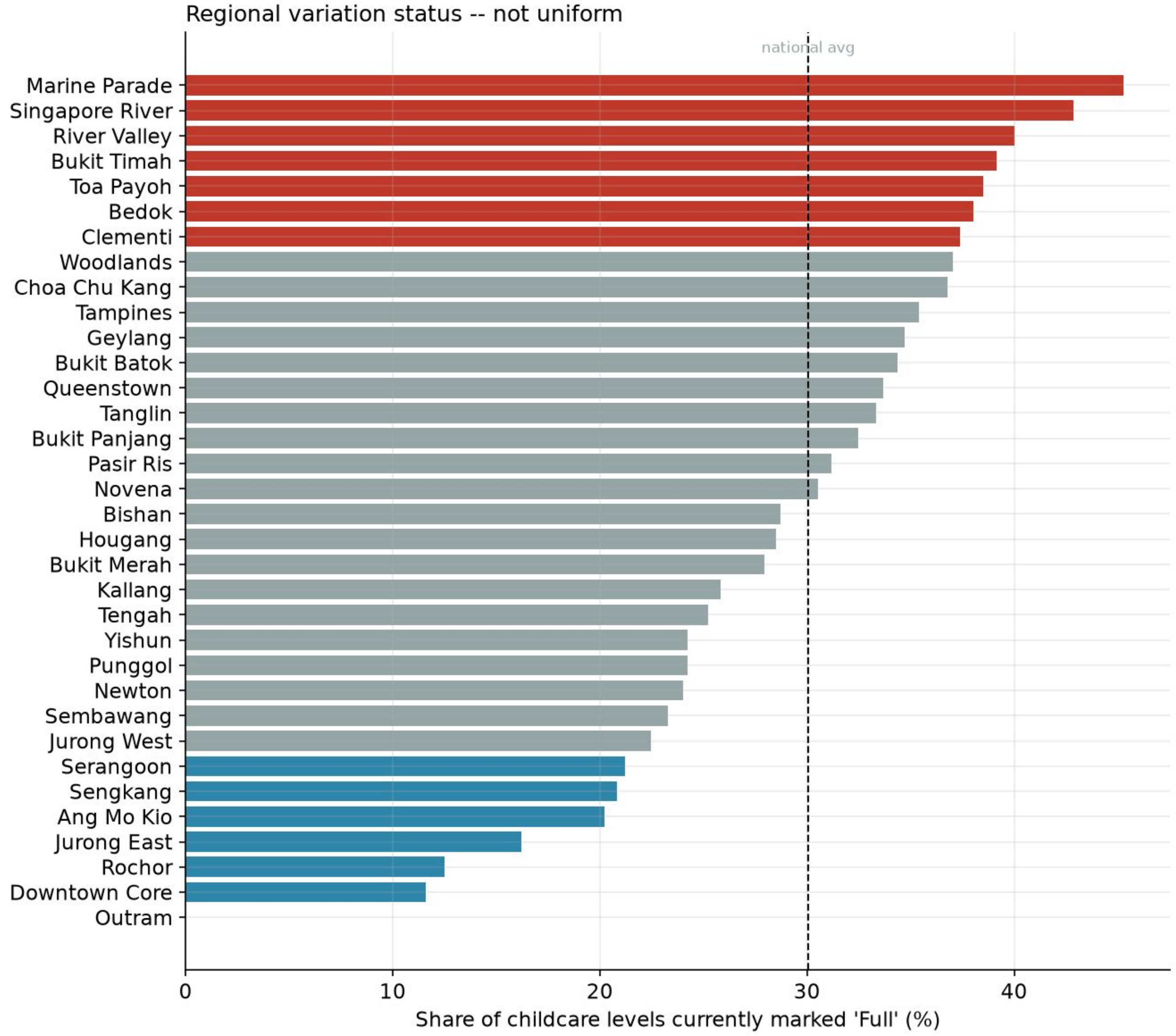
FRAMING THE QUESTION

Where are the Shortages?



1. The highest childcare occupancy is concentrated in mature, central estates such as Marine Parade, Singapore River, River Valley, Bukit Timah, and Toa Payoh.

2. Newer BTO towns such as Punggol, Tengah, Sengkang, and Bukit Batok are not the most capacity-constrained today, despite their projected demand growth.

3. These findings suggest different intervention strategies: expand capacity in fast-growing towns while addressing oversubscribed centres in mature estates through targeted expansion.

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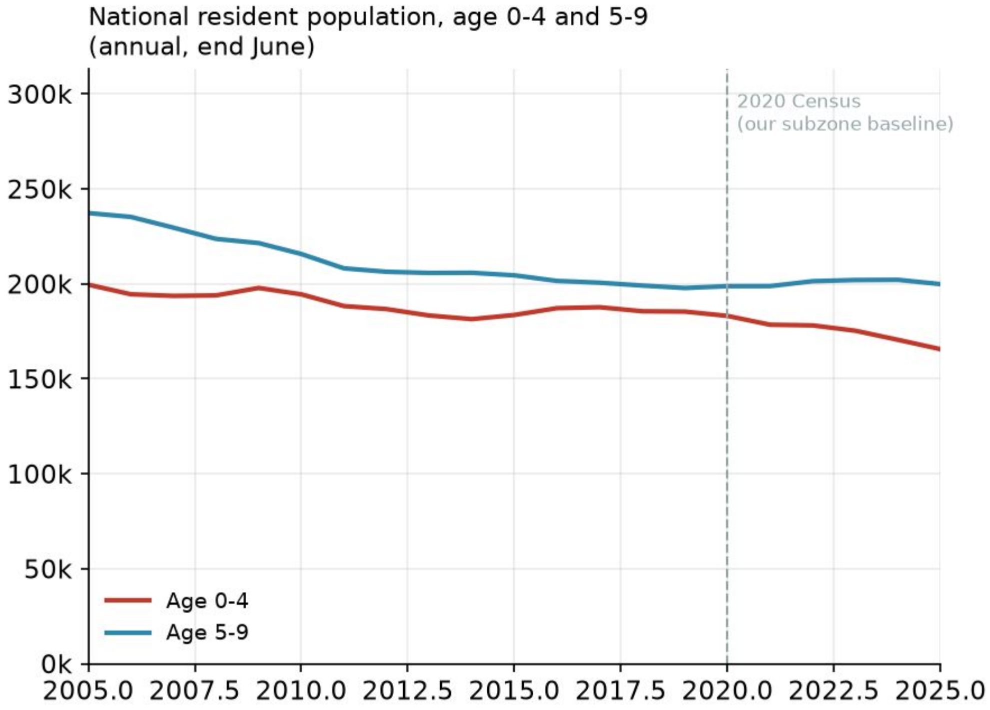
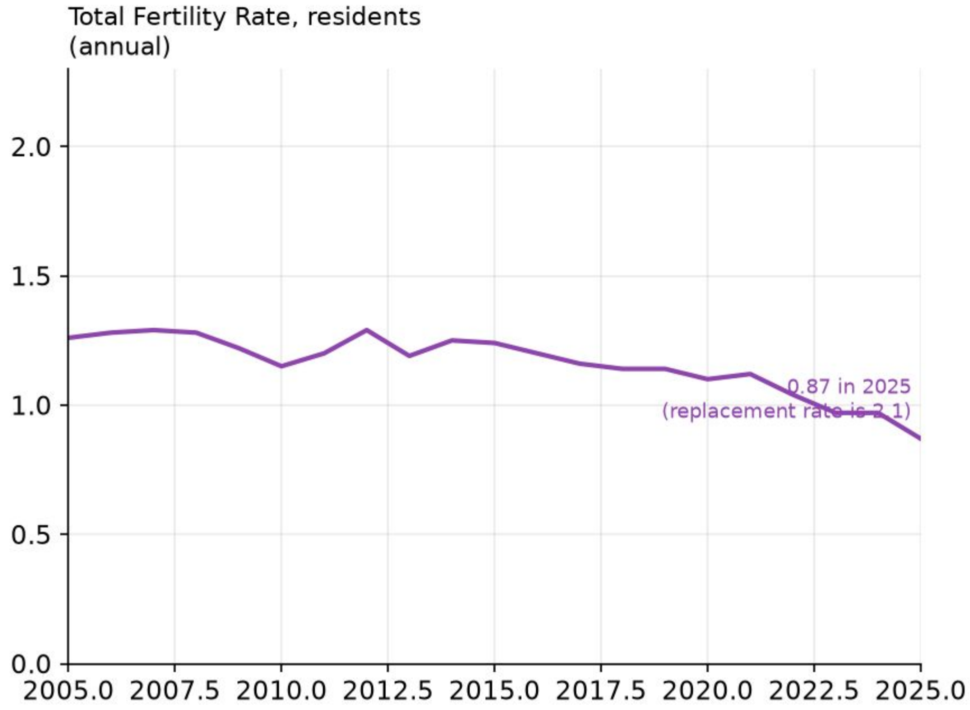
FRAMING THE QUESTION

Falling fertility will reduce demand in mature, non-BTO estates?





The national 0–4 child population has been declining in line with falling fertility rates, providing the baseline for understanding future childcare demand across subzones.

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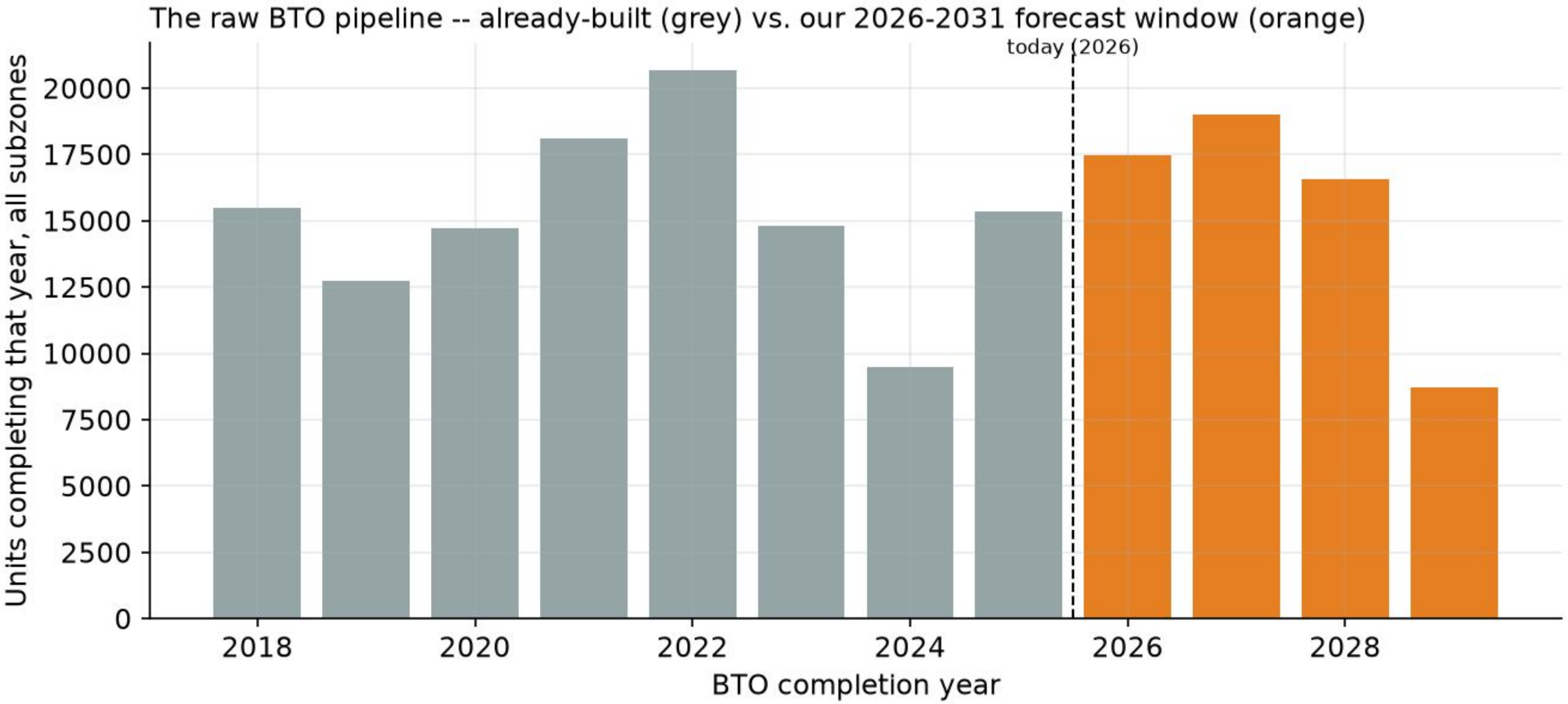
FRAMING THE QUESTION

BTO Forecast and its impact to Childcare Demand?

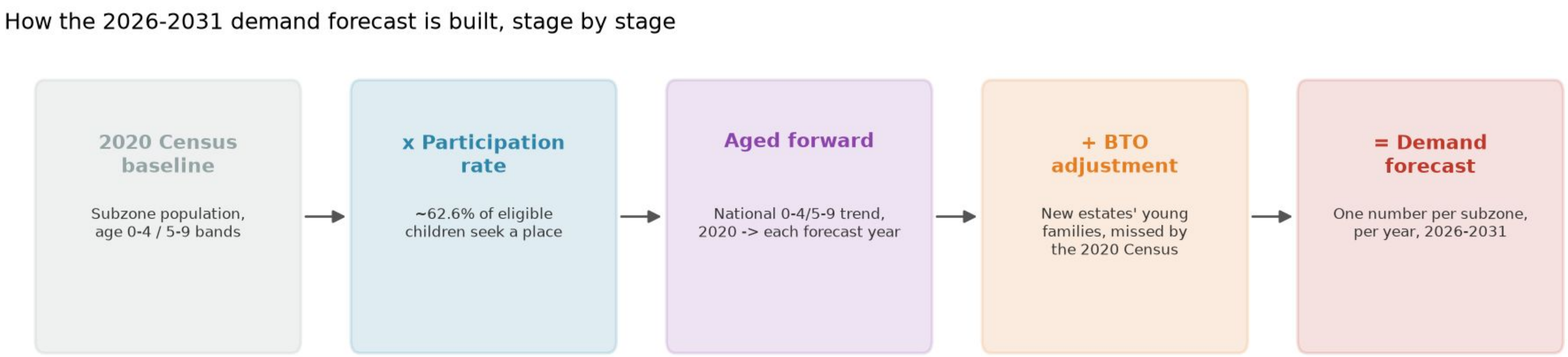


1. 5.13% of Singapore's population falls within the 18-month to 6-year childcare age group (2020 Census).
2. Applying this to a 3.2-person average BTO household gives a baseline of 0.164 childcare-age children per BTO unit.
3. A 1.7× uplift is applied to reflect that BTO buyers are predominantly young, first-time families.
4. Final assumption: **0.28** childcare-age children per BTO unit, used throughout the demand forecast.

66% of the BTO pipeline (121,419 units) has already been completed, contributing an estimated ~34,000 additional childcare-age children beyond the 2020 Census baseline and justifying the BTO adjustment.

The remaining 61,840 units (2026–2031) are projected to add ~17,300 children, with demand phased in over time as projects are completed and occupied.

Forecast Modelling



1. Estimate the eligible childcare population (18 months–6 years) by assuming an even age distribution within Census age bands: 18 months–4 years = 3.5/5 (70%) of the 0–4 population, and 5–6 years = 2/5 (40%) of the 5–9 population.

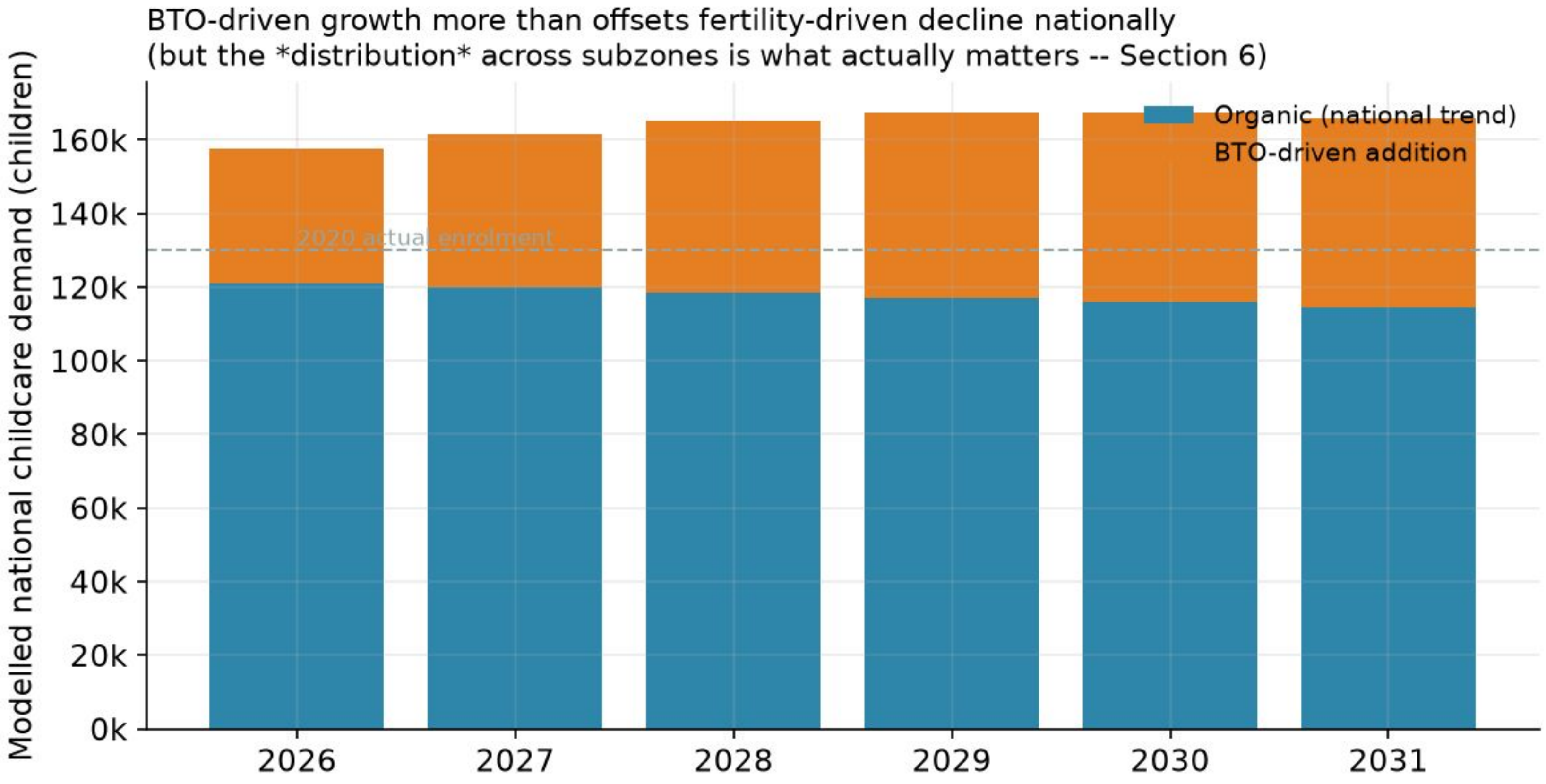
2. Estimate childcare demand by applying the 2020 national participation rate of 66%, calculated as 96,102 enrolled children ÷ 145,289 estimated eligible children, to convert the eligible population into expected enrolment.

3. Project future demand by first aging the baseline population using national child population trends, then adding BTO-driven demand based on 0.28 childcare-age children per completed BTO unit, phased in according to project completion and occupancy.

4. Output: annual childcare demand estimates for every subzone from 2026–2031, which are compared against existing childcare supply to identify future shortages.

Forecast Modelling

Aging forward + BTO: how national demand actually moves



1. **Organic**: national 0-4/5-9 trend aged forward from 2020 — tracking the fertility decline.
2. **+ BTO-driven**: each project phases in gradually — half-weight in its completion year, full from the year after. Not switched on overnight.
3. **Net**: BTO growth more than offsets the fertility decline — a modest +5% national rise, 2026→2031

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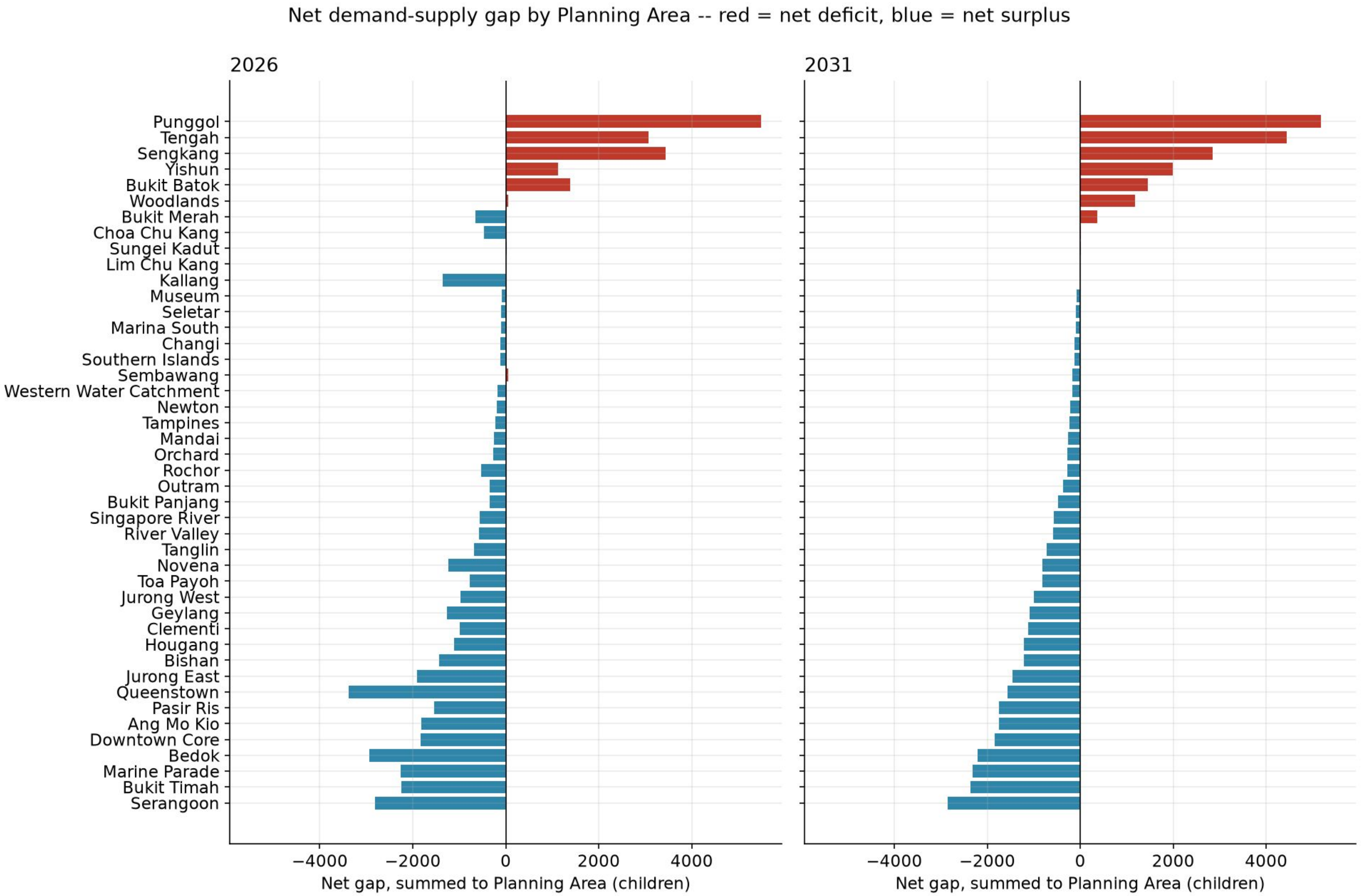
Gap Analysis & Prioritization



1. **Only 10 of 44 Planning Areas are projected to face a net childcare deficit by 2031**, compared with **89 deficit subzones**, as local surpluses and deficits offset each other when aggregated.

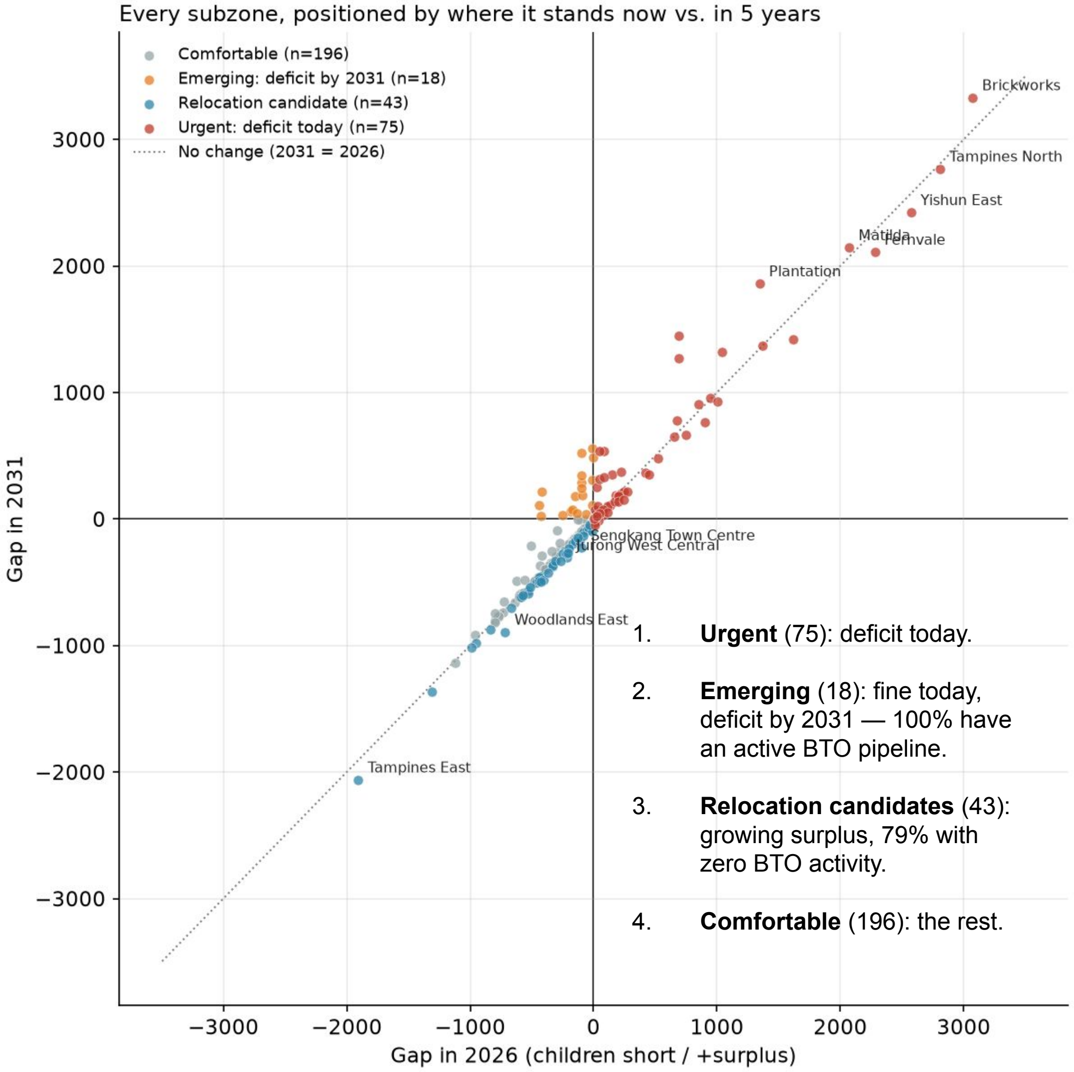
2. **This shows that childcare shortages are primarily a subzone-level issue**, with underserved subzones often hidden within Planning Areas that appear adequately served overall.

3. **Punggol records the largest projected net deficit (+5,182 children)**, consistent with its large BTO pipeline and rapidly growing young-family population.

4. **Serangoon records the largest projected net surplus (−2,852 children)**, making it a strong candidate for future capacity optimisation or relocation as demand shifts over time.

GAPS

Gap Analysis & Prioritization



1. The chart compares each subzone's childcare gap today (x-axis) against its projected gap in 2031 (y-axis).
2. Subzones above the diagonal line are projected to see worsening shortages, while those below are expected to improve over time.
3. Existing deficit subzones (red) largely remain in deficit or deteriorate further, indicating persistent capacity shortfalls.
4. Emerging deficit subzones (orange) are currently well served but are projected to fall into deficit by 2031, largely driven by upcoming BTO completions, while relocation candidates (green) become increasingly surplus over time.

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List 1 — build or relocate priority

subzone	planning_area	category	Centres Today	2026 Gap (children, + short / - surplus)	2031 Gap (children, + short / - surplus)	centres_needed_2031	bto_share_2031
Brickworks	Bukit Batok	Urgent: deficit today	12	3073	3329	34	72%
Tampines North	Tampines	Urgent: deficit today	1	2808	2766	28	86%
Yishun East	Yishun	Urgent: deficit today	25	2575	2422	25	47%
Matilda	Punggol	Urgent: deficit today	14	2070	2144	22	11%
Fernvale	Sengkang	Urgent: deficit today	23	2281	2108	22	31%
Plantation	Tengah	Urgent: deficit today	4	1346	1859	19	100%
Lower Seletar	Yishun	Urgent: deficit today	6	689	1447	15	41%
Waterway East	Punggol	Urgent: deficit today	24	1615	1419	15	17%
Northshore	Punggol	Urgent: deficit today	16	1367	1366	14	100%
Park	Tengah	Urgent: deficit today	5	1039	1317	14	100%
Garden	Tengah	Urgent: deficit today	5	689	1271	13	100%
Keat Hong	Choa Chu Kang	Urgent: deficit today	14	944	955	10	19%
Sembawang East	Sembawang	Urgent: deficit today	9	1007	931	10	49%
Woodlands West	Woodlands	Urgent: deficit today	8	853	909	10	48%
Woodlands South	Woodlands	Urgent: deficit today	11	677	778	8	25%

List 2 — Relocation Candidates

subzone	planning_area	Centres Today	2026 Gap (children, + short / - surplus)	2031 Gap (children, + short / - surplus)	gap_trend
Woodlands East	Woodlands	42	-717	-894	-177
Tampines East	Tampines	54	-1906	-2062	-156
Sengkang Town Centre	Sengkang	24	-100	-228	-129
Jurong West Central	Jurong West	21	-217	-306	-89
Punggol Field	Punggol	18	-12	-98	-85
Yunnan	Jurong West	22	-406	-483	-78
Yishun South	Yishun	16	-424	-493	-69
Sembawang Central	Sembawang	19	-266	-332	-65
Yishun West	Yishun	14	-207	-270	-63
Pasir Ris Drive	Pasir Ris	17	-526	-588	-62
Peng Siang	Choa Chu Kang	12	-114	-173	-59
Midview	Woodlands	14	-369	-426	-57
Fajar	Bukit Panjang	11	-82	-136	-55
Frankel	Bedok	24	-1313	-1365	-51
Simei	Tampines	14	-440	-489	-48

PRIORITIZATION

Different Outlook when considering Service Type

Blind spot: Every number so far assumes "1 centre = 100 kids, full 18mo-6y range."

subzone	planning_area	Centres Today	2026 Gap (children, + short / - surplus)	Younger-Band Gap (18mo-4y)	Older-Band Gap (5-6y)	masked_driver
Bedok North	Bedok	27.0	-273.864598	7.573243	-39.771174	Younger (18mo-4y) short
Sembawang Central	Sembawang	19.0	-266.249063	1.922918	-59.838648	Younger (18mo-4y) short
Jurong West Central	Jurong West	21.0	-216.644383	-86.898043	86.920327	Older (5-6y) short
Clementi Central	Clementi	6.0	-151.857778	-114.882633	46.358188	Older (5-6y) short
Bangkit	Bukit Panjang	6.0	-151.681581	-95.767063	27.418816	Older (5-6y) short
Townsville	Ang Mo Kio	6.0	-126.670791	-63.538620	3.534495	Older (5-6y) short
Pei Chun	Toa Payoh	3.0	-105.666159	-98.969759	26.636934	Older (5-6y) short
Sengkang Town Centre	Sengkang	24.0	-99.708425	79.994477	78.630431	Both bands short
Bukit Batok West	Bukit Batok	5.0	-82.148608	46.461380	-95.276655	Younger (18mo-4y) short
Fajar	Bukit Panjang	11.0	-81.827110	54.524109	-36.351219	Younger (18mo-4y) short

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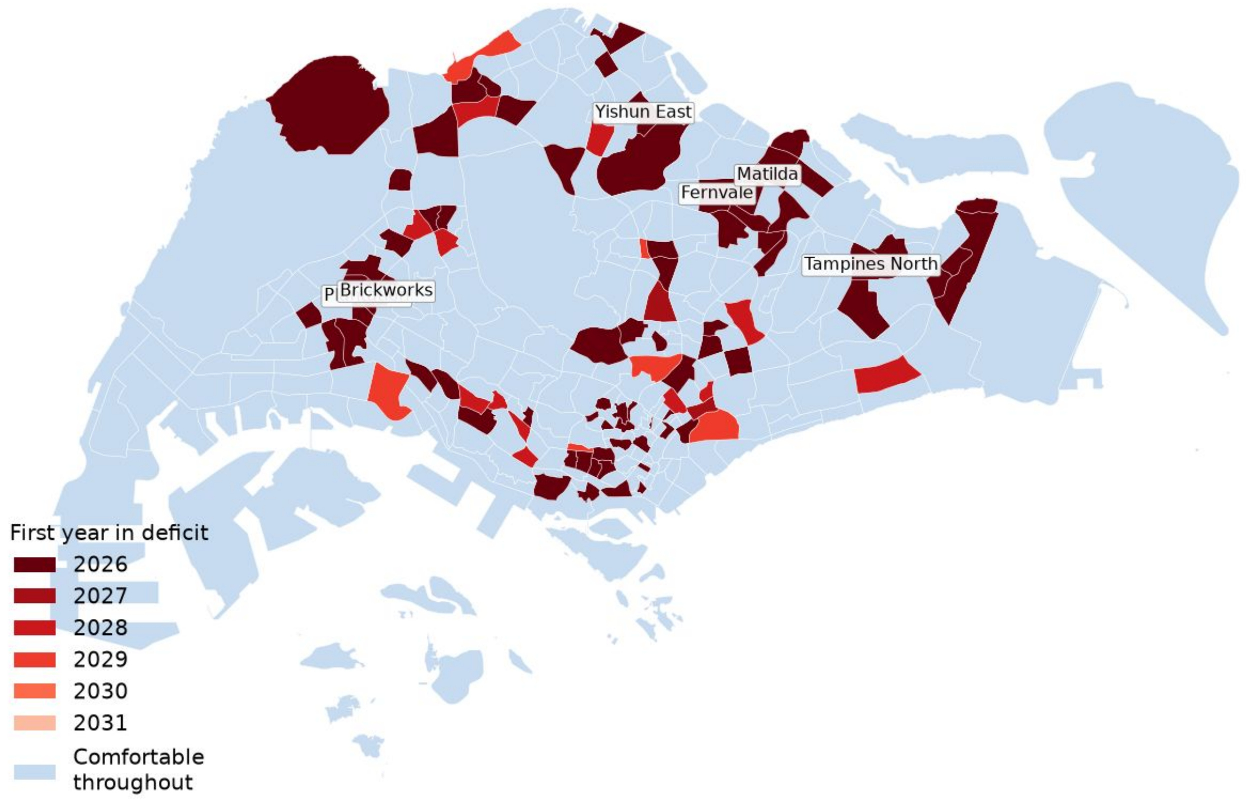
FORWARD LOOKING

Deficiency Roadmap Map

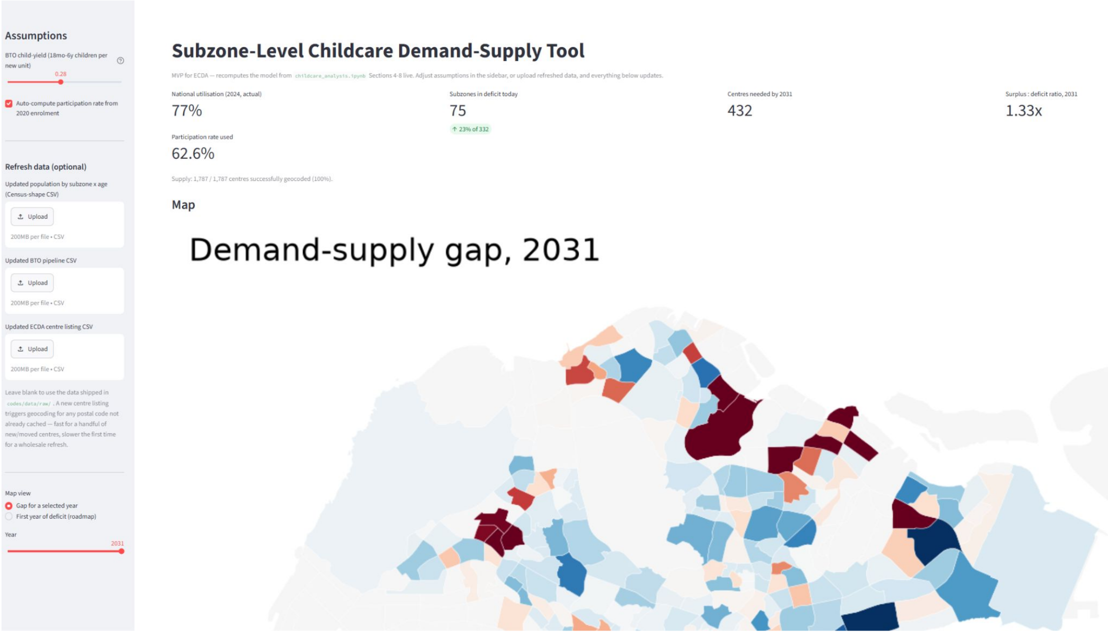
When each subzone first tips into deficit

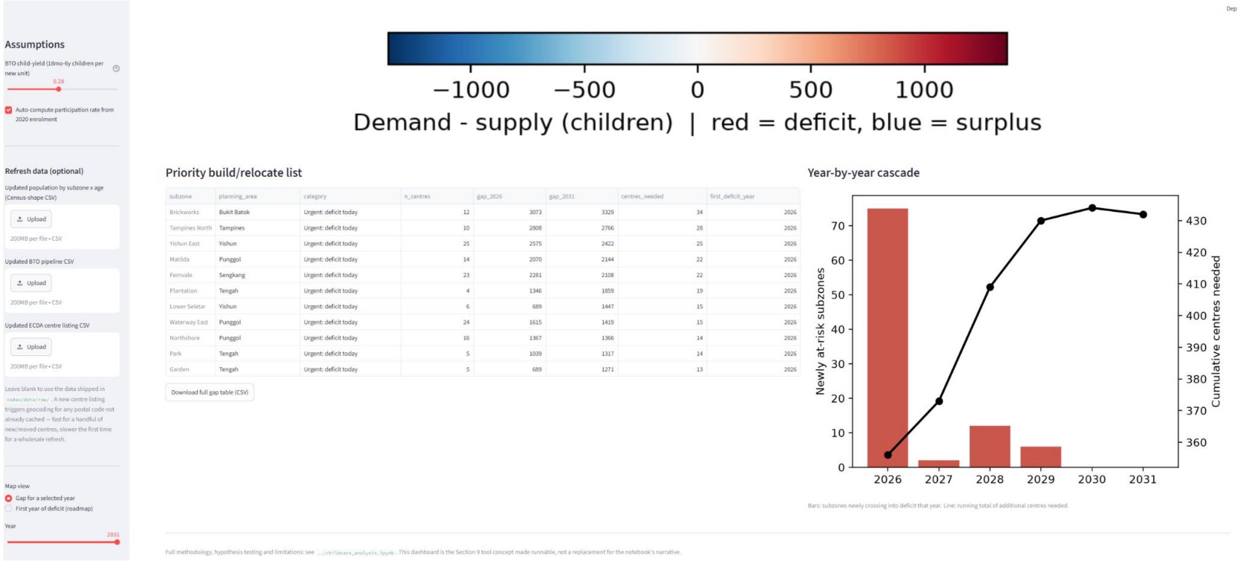


What the map shows: The darkest red areas represent subzones already facing childcare shortages today, concentrated across both mature estates and newer towns undergoing significant BTO development. Over time, additional BTO clusters gradually transition into deficit as new housing is completed and occupied. In contrast, the blue areas remain comfortably served throughout the forecast period, making them the strongest candidates for future capacity relocation rather than new centre construction.

Refer to codebase for robustness check

Forecast Tools





1. **Live demand–supply forecasting:** Instantly recalculates childcare gaps as assumptions (e.g. BTO child yield and participation rate) are adjusted, updating maps, KPIs, priority rankings, and charts in real time.
2. **Interactive scenario analysis:** Supports multiple map views (gap magnitude by forecast year or first year in deficit) and exports the resulting gap tables under the selected assumptions.
3. **Data refresh capability:** Allows users to upload updated population, BTO, or childcare centre datasets, with automatic recalculation and efficient re-geocoding of new centres.

How to run it:

From the section-2-case-study/codes/tool folder:

pip install -r requirements.txt

streamlit run app.py